Name: ________ Cohort/Batch: ________ Date: ________ Score: ________

HIVE PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Engineering

TOTAL: 10 QUESTIONS

Q1 What is Hive and what problem in Data Engineering does it solve? *Threat Model*

Q2 What are the core components or concepts in Hive? *Architecture*

Q3 How does Hive compare to its main alternatives? *Configuration*

Q4 How do you get started with Hive for a new project? *Hardening*

Q5 What configuration options most affect Hive's behavior in production? *SSO / IdP*

Q6 How does Hive handle reliability and high availability? *Observability*

Q7 What observability does Hive provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in Hive? *Compliance*

Q9 What security best practices apply when running Hive? *Testing*

Q10 What mistakes do teams commonly make when adopting Hive? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Hive is a tool or platform that

Mitigation

Hive is a tool or platform that automates or simplifies a specific concern in the Data Engineering domain, reducing manual effort and operational complexity for engineering teams.

A2 Hive has key building blocks that work

Primitives

Hive has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 Hive trades off simplicity against feature richness

Hardening

Hive trades off simplicity against feature richness differently than its alternatives. Choose Hive when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install Hive via its package manager or

Gotchas

Install Hive via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include concurrency

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 Hive provides HA through leader election, replication,

Observability

Hive provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 Hive exposes metrics (Prometheus endpoint or built-in

Automation

Hive exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from misconfigured

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run Hive with the principle of least

Assessment

Run Hive with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the documentation and

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.